Efficient $(1, s)$-Nucleus Decomposition on Billion-Scale Graphs

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Abstract

Extracting cohesive substructures from large graphs is a fundamental task in graph mining, with applications in community detection, dense-subgraph search, and role analysis. The $(1, s)$-*nucleus decomposition* is a higher-order generalization of k-core that ranks each vertex by the s-clique witnesses surrounding it (triangles for $s = 3$, 4-cliques for $s = 4$, and so on), giving sharper rankings than the standard k-core but at much higher cost. Existing studies compute $(1, s)$-nucleus inside a general (r, s) framework that encodes the witnesses with a Clique Path Index but maintains the index as mutable residual state during peeling. However, this mutability causes significant overhead in residual-index rewrites, memory consumption, and per-pop work, limiting scalability to large graphs and high s. We observe that the $(1, s)$ special case has a stronger invariant: vertex peeling never deletes edges, so every CPI path remains structurally valid throughout the peel. This invariant turns exact decomposition into counter maintenance over a static CPI, which yields SPIN, a direct static-path iteration, SPIN*, its incremental optimization, and a parallel builder for the shifted construction bottleneck. Experiments on 307 matched successful configurations show that SPIN* achieves a 13.50× geometric-mean peel-time speedup and a 44.76× maximum peel-time speedup over the mutable baseline, while reducing memory usage by a 1.55× geometric-mean ratio; the static-state pipeline also scales to a billion-edge graph for which the mutable baseline completes only at s=2.

CCS Concepts

• **Theory of computation** → **Graph algorithms analysis**; • **Information systems** → *Data mining*.

Keywords

nucleus decomposition, clique counting, cohesive subgraphs, hierarchy

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1 Introduction

Core decomposition ranks vertices by the amount of cohesive structure that still surrounds them after weak vertices are removed [6, 42, 49] and supports a wide range of graph-mining tasks — community detection [14, 18, 19, 21], dense-subgraph discovery [20], influential-vertex and role analysis [12, 25, 31], network visualization [51], and applications in social, biological, and security networks [3, 34, 35]. The classical k-core uses edges as the witness, but many of these tasks need a stronger witness because a group may pass an edge-degree threshold while containing few triangles, 4-cliques, or larger coordinated patterns [1, 13, 23]. The $(1, s)$-*nucleus decomposition* [40, 41] gives this vertex-centric higher-order ranking: for a fixed s, it assigns each vertex the largest value k for which the vertex belongs to a maximal region where every vertex participates in at least k internal s-cliques. For $s = 2$ it reduces to the k-core; for larger s it ranks vertices by progressively tighter clique-supported cohesion.

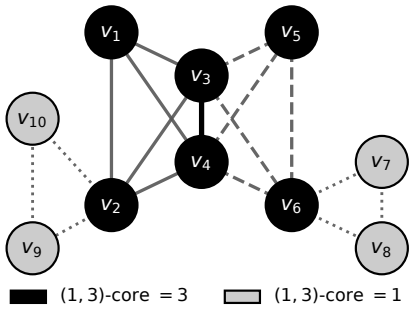

Figure 1: Running example of $(1, 3)$-nucleus decomposition.

Example 1.1. Figure 1 shows a 10-vertex graph at $s = 3$. The six vertices $v_1, \ldots, v_6$ belong to two triangle-rich K_4 blocks; each of them participates in at least three triangles supported by other vertices of the same value, so all six have $(1, 3)$-core value 3. The remaining four vertices $v_7, \ldots, v_{10}$ belong to a single pendant triangle, supporting only one triangle each, so all four have $(1, 3)$-core value 1. A formal definition is given in Section 2; we revisit this graph as a running example throughout the paper.

Challenges. Computing $(1, s)$-nucleus decomposition at high s is hard because the number of s-cliques can be enormous; the current state of the art avoids explicit enumeration with the Clique Path Index, or CPI [4]. Using the CPI efficiently for the vertex-centric case $r = 1$ raises four challenges.

Residual state during peeling. The general (r, s) algorithm maintains the CPI as mutable residual state, storing per-path vertex sets, a reverse vertex-to-path index, and a priority queue, and rewriting these structures every time a vertex is popped. At $r = 1$, peeling removes vertices but never deletes edges, so it is not obvious that the same residual maintenance is necessary.

Memory footprint of the residual index. The residual structures dominate memory: per-path vertex hash sets, the reverse vertex-to-path map, and the live Bron–Kerbosch recursion tree together pay

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constant-factor overhead per vertex–path incidence on top of the clique-encoding cost. A layout whose memory grows with the vertex–path incidence count Σ alone would let the achievable s on a fixed-memory machine be dictated by Σ rather than by hash-table overhead.

Construction as the new bottleneck. Once peeling becomes counter-only, the static index builder takes seconds to minutes while peeling takes milliseconds, so construction is the next bottleneck and must be parallelised without paying the synchronisation cost of mutable path rewrites.

Hierarchy emission from a mutating index. The clique-coherent nucleus hierarchy of Definition 2.3 witnesses s-connectivity through s-cliques contained in each super-level set, so emitting it requires access to the s-cliques themselves. If the CPI is mutated during peeling, those cliques are no longer available afterwards and the hierarchy needs either a second clique-enumeration pass or per-peel-level state, both of which can dominate total cost.

State-of-the-Art and Limitations. The current state of the art for general (r, s)-nucleus decomposition is CND [4], which maintains the CPI as mutable residual state and inherits four weaknesses when restricted to $r = 1$.

Mutable residual-index maintenance. CND rewrites path records and updates a hash-indexed reverse map at every peel step. These rewrites are a real cost rather than a constant factor: SPIN*, which replaces them with closed-form counter updates over a static index, delivers a 13.50× geometric-mean peel-time speedup over CND on 307 matched cells (Section 8).

High memory consumption. The residual structures that CND must keep mutable in the general (r, s) setting — per-path membership queries, reverse vertex-to-path lookup, and the live recursion tree used to rewrite paths — together cost roughly 88Σ bytes per incidence (Section 8, Table 3), close to an order of magnitude more than what a static encoding of the same witnesses requires. The geometric-mean memory ratio CND/SPIN* is 1.55× on the matched cells and 3.92× at maximum; on a billion-scale graph CND completes only at s=2 (using 400 GB of 503 GB) and exceeds the budget for every $s \geq 3$, while SPIN* finishes the full s sweep.

Construction as the residual bottleneck. Even after cheap peeling, CND pays the full construction cost; on `web-it-2004` (our highest CPI-build-cost benchmark) at $s = 3$, construction takes about 127 seconds and peeling takes 351 milliseconds, so the static index builder dominates total runtime by 360× on this configuration. Without a parallel builder, the end-to-end speedup over CND stays at 1.10× on the matched cells even though the peel-time speedup is 13.50×.

Hierarchy emission re-enumerates or replicates. Because CND edits the CPI during peeling, its s-clique witnesses are gone when the hierarchy is requested. The natural workaround — the general (r, s) level-DSU hierarchy specialised to r=1 — maintains one disjoint-set forest per core level during the peel and exceeds the available 503 GB on `wiki-Talk` at $s \geq 9$ where our pipeline finishes (Figure 9); even where it fits, it uses up to 43× more memory (gmean 13× over 24 matched cells) than emitting the same hierarchy from a preserved CPI.

Our Solution and Contributions. For $(1, s)$-nucleus decomposition we show that vertex peeling preserves the CPI structure: a set of vertices that formed a clique when the CPI was built remains a clique throughout the peel, and a path's only dynamic state is an integer counter of how many of its pivots remain. This invariant turns exact decomposition into counter arithmetic over a static incidence layout, which unlocks an incremental peel schedule and a parallel builder.

*Static-*CPI *invariant for* $(1, s)$. We prove that vertex peeling never rewrites the stored hold or pivot sets (Theorem 4.1) and give a closed-form expression for the support loss caused by each path event (Lemma 4.2). The dynamic state of a path collapses to a liveness bit and a single integer counter, so the per-peel residual rewrites disappear; the headline Exp-1 peel speedup over CND on 307 matched cells is 13.50× in the geometric mean.

Elimination of residual structures. A corollary of Theorem 4.1 is that three of CND's residual structures cease to carry information once the invariant is established: (i) the per-path vertex hash sets, because membership in a path is already determined by the path's fixed hold/pivot sets; (ii) the reverse vertex-to-path map, because the vertex-to-path relation is fixed at construction and never edited; and (iii) the mutable Bron–Kerbosch recursion tree, because no path is ever rewritten during peeling. This collapses the per-incidence space cost from $\approx 88\Sigma$ bytes to $\approx 8\Sigma$ bytes (Table 3); the measured geometric-mean memory ratio CND/SPIN* is 1.55× over 307 matched cells, and on a billion-scale graph SPIN* decomposes the full s sweep while CND OOMs for every $s \geq 3$.

*Parallel static-*CPI *construction.* Because the static-CPI output is invariant under the order in which seed vertices are processed and contains no mutable shared state, seed enumeration partitions into independent subproblems. ParaBuild (Algorithm 5) constructs the index by running these subproblems in parallel and combining their outputs by a deterministic merge whose result does not depend on the schedule; the empirical scaling reaches 23.4× at T=24 threads.

Direct and incremental solvers over the static index. We formulate SPIN as the direct threshold-support iteration on CPI witness counts (Algorithm 3) and SPIN* as its incremental optimization (Algorithm 4) that computes the same fixed point in peel order without repeated full-graph passes (Lemma 5.5). SPIN* delivers a 30.4× geometric-mean peel-time speedup over SPIN on 272 matched cells.

Core-value hierarchy from the static CPI. After peeling, each CPI leaf can be treated as a hyperedge born at the minimum core value among its stored vertices. A descending union–find scan over the leaves recovers the s-clique-connected super-level join tree in $O(\Sigma \log \Sigma)$ time without re-enumerating s-cliques (Algorithm 6); the hierarchy is a byproduct of preserving the CPI rather than a separate decomposition problem.

Comprehensive experimental evaluation. We compare against CND on ten real-world graphs (307 matched configurations, headline peel speedup 13.50× gmean / 44.76× max, memory-usage ratio 1.55× gmean), report phase behaviour and parallel construction, push the pipeline to a billion-scale graph (where CND only completes at s=2), and use case studies to evaluate whether the $r = 1$ specialization is practically useful.

2 Preliminaries

Graph notation. We work with a simple undirected graph $G = (V, E)$. For a vertex v, $N_G(v) = \{u \in V \mid (u, v) \in E\}$ is its neighbor set and $\deg_G(v) = |N_G(v)|$ is its degree. For $X \subseteq V$, $G[X]$ is the

subgraph induced by X. The implementation stores G in CSR form with an `offsets` array and an `edges` array.

Definition 2.1 (k-Clique)**.** A *k-clique* is a vertex set $C \subseteq V$ with $|C| = k$ such that every two distinct vertices in C are adjacent.

Definition 2.2 (s-Clique Support)**.** Given $s \geq 2$, the *s-clique support* of a vertex v in G is

$$\deg^{(s)}(v) \;=\; \big|\{S \subseteq V \mid v \in S,\ |S| = s,\ S \text{ is a clique in } G\}\big|.$$

$\deg^{(s)}(v)$ is evaluated inside the current induced subgraph when a peel level is being defined; the subscript is dropped when the graph is clear.

Definition 2.3 (k-$(1,s)$-Nucleus)**.** Let $X \subseteq V$. The induced subgraph $G[X]$ is a *k-$(1,s)$-nucleus* if:

(1) every vertex in X belongs to at least k s-cliques contained in $G[X]$;
(2) $G[X]$ is *s-connected*: for any two vertices $u, v \in X$, there is a sequence $u = v_1, \ldots, v_t = v$ such that each pair $\{v_i, v_{i+1}\}$ is contained in an s-clique of $G[X]$;
(3) $G[X]$ is maximal with respect to adding vertices while preserving the first two conditions.

Definition 2.4 (Core Number)**.** Given $s \geq 2$, the *$(1,s)$-nucleus core number* of a vertex v, denoted $\kappa_s(v)$, is the largest k for which v belongs to some k-$(1,s)$-nucleus of G.

The Clique Path Index. The Clique Path Index, or CPI, compactly represents s-cliques without listing them one by one [4].

Definition 2.5 (CPI Path)**.** A CPI *path* is a pair $\mathcal{P} = (V_h(\mathcal{P}), V_p(\mathcal{P}))$ of disjoint vertex sets such that $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ is a clique in G and $|V_h(\mathcal{P})| \leq s \leq |V_h(\mathcal{P})| + |V_p(\mathcal{P})|$. The set $V_h(\mathcal{P})$ is the *hold set*: every encoded s-clique contains all of these vertices. The set $V_p(\mathcal{P})$ is the *pivot set*: an encoded s-clique chooses vertices from this pool. The *pivot requirement* of $\mathcal{P}$ is $\eta_{\mathcal{P}} = s - |V_h(\mathcal{P})|$, the number of pivot vertices needed to complete the hold set into an s-clique. The set of s-cliques encoded by $\mathcal{P}$ is

$$C(\mathcal{P}) = \big\{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}),\ |P'| = \eta_{\mathcal{P}}\big\}.$$

Its cardinality is $\binom{|V_p(\mathcal{P})|}{\eta_{\mathcal{P}}}$.

The CPI is a set $\mathcal{P}$ of paths such that every s-clique is encoded by exactly one path. We write

$$\Sigma = \sum_{\mathcal{P} \in \mathcal{P}} \big(|V_h(\mathcal{P})| + |V_p(\mathcal{P})|\big)$$

for the vertex–path incidence count. We also write $\deg_\Sigma(v) = |\{\mathcal{P} \mid v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})\}|$, $d_{\max} = \max_v \deg_\Sigma(v)$, and $\kappa_{\max} = \max_v \deg^{(s)}(v)$. The CPI is built by a degeneracy-ordered Bron–Kerbosch search with Tomita pivoting [4, 10, 16, 45]; we reproduce the construction here as Algorithm 1 because the rest of the paper depends on its leaf shape.

Algorithm 1 computes a degeneracy ordering of G and orients edges from lower to higher rank so each s-clique has a unique smallest-rank owner (Line 1); for every vertex v_i, ExpandPath is invoked with hold set $\{v_i\}$ and forward-neighbour candidate set $N^+(v_i)$ (Lines 3-4). A recursive call returns immediately when the hold set already exceeds s (Line 7), and emits a path when the candidate set is empty and the hold-plus-pivot size meets s (Lines 8-9). Otherwise the Tomita pivot u_{piv} is the vertex of P with the most forward neighbours inside P (Line 10), and every non-neighbour of u_{piv} is recursively expanded: branching on the pivot itself extends the pivot set, branching on any other vertex extends the hold set (Lines 11-14).

Algorithm 1: BuildCPI(G, s) — standard CPI construction [4].

```
Input: Undirected graph G = (V, E), clique size s.
Output: Clique Path Index P of G for target size s.
1  Compute a degeneracy ordering (v_1, ..., v_n) of G and orient every edge
     from lower to higher rank, giving N^+(·);
2  P ← ∅;
3  foreach i ← 1 to n do
4      ExpandPath(N^+(v_i), {v_i}, ∅) ;          // one root per vertex
5  return P   Function ExpandPath(P, V_h, V_p):
6      if |V_h| > s then return;
7      if P = ∅ then
8          if |V_h| + |V_p| ≥ s then append the path (V_h, V_p) to P ;
9          return
10     u_piv ← arg max_{x∈P} |P ∩ N^+(x)| ;          // Tomita pivot
11     foreach v ∈ P \ N(u_piv) do
12         P ← P \ {v};
13         if v = u_piv then
14             ExpandPath(P ∩ N(v), V_h, V_p ∪ {v});
15         else
16             ExpandPath(P ∩ N(v), V_h ∪ {v}, V_p);
```

Lemma 2.6 (Support Formula [4])**.** *For every $v \in V$,*

$$\deg^{(s)}(v) = \sum_{\mathcal{P}\,:\,v \in V_h(\mathcal{P})} \binom{|V_p(\mathcal{P})|}{\eta_{\mathcal{P}}} + \sum_{\mathcal{P}\,:\,v \in V_p(\mathcal{P})} \binom{|V_p(\mathcal{P})| - 1}{\eta_{\mathcal{P}} - 1}.$$

Problem statement. Given G and $s \geq 2$, compute $\kappa_s(v)$ for every $v \in V$; the nucleus hierarchy is constructed afterwards (Section 7).

3 The Mutable-CPI Baseline

CND [4] is the state-of-the-art exact algorithm for general (r,s)-nucleus decomposition. The remainder of this section walks through its peeling loop, the residual structures it maintains, and the operations this paper specialises away when restricted to $r = 1$.

The peeling loop. CND follows the standard core-decomposition template [6]: it maintains the residual support of every alive r-clique in a priority queue, repeatedly pops the r-clique of minimum residual support, assigns it the current peel level as its core value, and propagates the support loss to the remaining alive r-cliques. At $r = 1$ the queue items are vertices, so each iteration pops a vertex v, records $\kappa_s(v)$, and decrements the residual support of every vertex that loses an s-clique witness because of v's removal. The non-trivial work happens during this support-loss propagation, because the CPI that encodes the witnesses must be brought into a state consistent with v being gone before the next pop.

Residual state. To keep the witness counts consistent across peels, CND maintains the CPI as mutable residual state. For each path $\mathcal{P}$ it stores the hold and pivot vertex sets as hash sets so that the test "does v belong to this path?" is constant time. A separate reverse map sends every vertex u to the list of paths that contain it, so that when u is popped the algorithm can find the affected paths without scanning all of $\mathcal{P}$. A heap (or bucket queue) keeps every alive vertex ordered by current residual support, with decrease-key operations for the support drops triggered at each peel event.

Finally, the Bron–Kerbosch recursion tree that produced the CPI is kept mutable: when a peel event invalidates a path, the algorithm walks into the tree and rewrites or removes the affected nodes.

Per-peel work. A single peel of vertex v costs:

- a reverse-map lookup of v's path list, then for every such path $\mathcal{P}$ a constant-time hash-set query to determine whether v is a hold or a pivot;
- a path rewrite: if v is a hold, the path is deleted from the recursion tree and its reverse-map entries for all other vertices in $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ are erased; if v is a pivot, the path's pivot set shrinks by one and the reverse-map entry for v on this path is erased;
- a support recompute for every path-mate u using a freshly enumerated set of s-cliques that survive the rewrite, followed by a decrease-key on u in the heap.

The dominant components are the path rewrites and the reverse-map erases; together they scale with the number of vertex–path incidences touched by the event, which is what gives CND its observed cost profile.

Why this state is needed in the general case. The mutable design is not over-engineering for the general (r, s) setting. When the peeled object is an edge ($r = 2$) or a higher-order clique ($r \geq 3$), the removal can destroy edges among vertices that remain alive; some of the cliques that a path encodes therefore cease to exist as cliques. In that setting, the stored hold and pivot sets of $\mathcal{P}$ no longer define a valid encoded clique family, so the path must be split, shrunk, or deleted by walking back into the recursion tree. The reverse map and the per-path hash sets exist precisely to support that surgery efficiently.

Why this state is wasteful at $r = 1$. The $(1, s)$ case has a stronger invariant: vertex peeling removes vertices but does not delete edges, so a set of vertices that formed a clique when the CPI was built remains a clique throughout the peel (we prove this in Section 4). A path's encoded clique family therefore stays structurally valid; only the question of whether a particular vertex of $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ is still alive changes. This means the path rewrites, the reverse-map erases, and the recursion-tree surgery in the per-peel cost above are all unnecessary at $r = 1$ — they maintain a mutable residual state that is logically equivalent to a static incidence layout plus a single integer counter per path. The next four sections cash in this observation: Section 4 states the static-CPI invariant in counter form; Section 5 derives SPIN and SPIN* over the resulting static index; Section 6 parallelises construction once peeling becomes counter-only; and Section 7 extracts the nucleus hierarchy from the preserved CPI without any clique re-enumeration.

4 Static CPI for $r = 1$

We prove the central invariant of the paper: at $r = 1$, vertex peeling never deletes an edge among the surviving vertices, so the stored clique $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ of every path remains structurally valid throughout the peel and the residual state that CND maintains (Section 3) is unnecessary.

Static structure and dynamic counters. Recall from Section 2 that a path stores the hold set $V_h(\mathcal{P})$, the pivot set $V_p(\mathcal{P})$, and the pivot requirement $\eta_{\mathcal{P}} = s - |V_h(\mathcal{P})|$; during peeling none of these three objects is edited. The dynamic state is only a liveness bit and an integer $p_{\mathcal{P}}$, the number of pivots in $V_p(\mathcal{P})$ that have not yet been peeled. If a peeled vertex is a hold, every encoded s-clique contains that vertex, so the path contributes nothing after the event; if a peeled vertex is a pivot, the path remains the same symbolic object and $p_{\mathcal{P}}$ decreases by one. The next theorem records this counter form: a peel event either kills a path contribution or reduces an effective pivot count.

Theorem 4.1 (Vertex Removal on CPI, Counter Form)**.** *Let $\mathcal{P}$ be a path of the* CPI *of G, and let $v \in V$ be removed from G.*

(1) If $v \in V_h(\mathcal{P})$, $\mathcal{P}$ encodes no surviving s-clique.

(2) If $v \in V_p(\mathcal{P})$, the surviving s-cliques are encoded by the same $V_h(\mathcal{P})$ together with $V_p(\mathcal{P}) \setminus \{v\}$. Their count depends only on $|V_p(\mathcal{P})|$. A counter $p_{\mathcal{P}} = |V_p(\mathcal{P})|$ decremented by one is therefore support-equivalent to deleting v from $V_p(\mathcal{P})$. If $p_{\mathcal{P}} - 1 < \eta_{\mathcal{P}}$, no surviving s-clique is encoded.

(3) Otherwise, $\mathcal{P}$ is unaffected.

The dynamic state of $\mathcal{P}$ during peeling reduces to an integer counter $p_{\mathcal{P}}$, the fixed pivot requirement $\eta_{\mathcal{P}}$, and a liveness bit. The stored sets $V_h(\mathcal{P})$ and $V_p(\mathcal{P})$ are never modified.

Proof. $\mathcal{P}$ encodes $C(\mathcal{P}) = \{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}),\ |P'| = \eta_{\mathcal{P}}\}$. If $v \in V_h(\mathcal{P})$, every encoded clique contains v and none survives. If $v \in V_p(\mathcal{P})$, the surviving cliques are $\{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}) \setminus \{v\},\ |P'| = \eta_{\mathcal{P}}\}$, a family of size $\binom{|V_p(\mathcal{P})|-1}{\eta_{\mathcal{P}}}$ that depends on $V_p(\mathcal{P})$ only through its cardinality; by Lemma 2.6 every support contribution of the path depends only on this cardinality, so the counter update $p_{\mathcal{P}} \leftarrow p_{\mathcal{P}} - 1$ is support-equivalent to physical deletion, and the family is empty iff $p_{\mathcal{P}} - 1 < \eta_{\mathcal{P}}$. If $v \notin V_h(\mathcal{P}) \cup V_p(\mathcal{P})$, no encoded clique mentions v and the family is unchanged. □ □

Boundary of the invariant. The theorem is specific to $r = 1$. When $r \geq 2$, a peel event removes an edge or a higher-order clique from the residual object. That event can break the clique represented by $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$. In that setting, a path may need to be split, shrunk, or rebuilt. The mutable CPI machinery of [4] remains the appropriate general tool, and this paper does not subsume it.

Support loss from one path event. Theorem 4.1 leaves only cardinality changes. The support lost by any other live vertex on the path is therefore a binomial difference.

Lemma 4.2 (Support Loss for a Path Event)**.** *Let $\mathcal{P}$ be a live path with $p \geq \eta$ before an event that removes $d \geq 1$ live pivots from $\mathcal{P}$. For every other alive vertex u incident to $\mathcal{P}$:*

$$\Delta_{hold}(p, \eta, d) = \binom{p}{\eta} - \binom{p-d}{\eta} \qquad \text{if } u \in V_h(\mathcal{P}),$$

$$\Delta_{pivot}(p, \eta, d) = \binom{p-1}{\eta-1} - \binom{p-d-1}{\eta-1} \qquad \text{if } u \in V_p(\mathcal{P}).$$

A path-killing event, caused by a hold peel or by $p - d < \eta$, is the limiting case in which the entire contribution vanishes: $\binom{p}{\eta}$ for surviving holds and $\binom{p-1}{\eta-1}$ for surviving pivots.

Proof. By Lemma 2.6, a hold u contributes $\binom{p}{\eta}$ before the event and $\binom{p-d}{\eta}$ after; a pivot u occupies one slot and contributes $\binom{p-1}{\eta-1}$ before and $\binom{p-d-1}{\eta-1}$ after. Subtracting after from before gives the two formulas. □ □

Table 1: CPI **paths for the running example at** $s = 3$.

Path	V_h	V_p	$\eta = s - \lvert V_h \rvert$	$\binom{\lvert V_p \rvert}{\eta}$
$\mathcal{P}_1$	$\{v_7\}$	$\{v_6, v_8\}$	2	1
$\mathcal{P}_2$	$\{v_9\}$	$\{v_2, v_{10}\}$	2	1
$\mathcal{P}_3$	$\{v_1\}$	$\{v_2, v_3, v_4\}$	2	3
$\mathcal{P}_4$	$\{v_2\}$	$\{v_3, v_4\}$	2	1
$\mathcal{P}_5$	$\{v_3\}$	$\{v_4, v_5, v_6\}$	2	3
$\mathcal{P}_6$	$\{v_4\}$	$\{v_5, v_6\}$	2	1
Total				**10**

Running example. Figure 1 shows the running example at $s = 3$: two K_4 blocks share edge (v_3, v_4), and two pendant triangles attach at v_2 and v_6. The six CPI paths in Table 1 encode the ten triangle witnesses through stored V_h, V_p sets and the binomial count $\binom{\lvert V_p \rvert}{s-\lvert V_h \rvert}$. The $(1, 3)$-core values fall into two bands: $\kappa_3(v_i) = 3$ for $i \in \{1, \ldots, 6\}$ (the two K_4 blocks) and $\kappa_3(v_i) = 1$ for $i \in \{7, \ldots, 10\}$ (the pendant triangles); this is the partition shown by the node shading in Figure 1. The example exercises both events of Theorem 4.1: a path dies when a hold vertex is removed, and a path's pivot counter shrinks when a pivot vertex is removed.

Example 4.3 (Peel event walk-through). Consider the path $\mathcal{P}_3 = (\{v_1\}, \{v_2, v_3, v_4\})$ in Table 1, with $p = 3$ and $\eta = 2$. Peeling v_2 as a pivot triggers Lemma 4.2 with $d = 1$. The surviving hold v_1 loses $\binom{3}{2} - \binom{2}{2} = 2$ from its support. The surviving pivots v_3 and v_4 each lose $\binom{2}{1} - \binom{1}{1} = 1$. The stored set $V_p(\mathcal{P}_3)$ is not modified. Only $p_{\mathcal{P}_3}$ changes from 3 to 2.

5 Computing Core Values over Static CPI

The static-CPI invariant of Section 4 becomes a concrete algorithm for κ_s through a one-vertex update rule built directly on clique-witness counts (Section 5.1, Algorithm 2). Two solvers share the same update: SPIN applies it by repeated full passes (Section 5.2, Algorithm 3); SPIN* schedules it in peel order so unaffected vertices are not revisited (Section 5.3, Algorithm 4), and Lemma 5.5 proves the two reach the same fixed point.

5.1 Vertex updates from static witnesses

The static-CPI invariant counts witnesses without rebuilding paths; for core values we additionally need a rule for lowering a vertex value from the witnesses that remain. At level k, the vertex needs at least k encoded clique witnesses whose other vertices can also sustain level k. For a tentative value vector, define the one-vertex update

$$\Phi_h(v) = \max\Big\{k \geq 0 \,\Big|\, \sum_{\mathcal{P} \ni v} W_v(\mathcal{P}, h, k) \geq k\Big\}.$$

Here $W_v(\mathcal{P}, h, k)$ is the number of encoded s-cliques on path $\mathcal{P}$ through v whose other vertices have value at least k, and the sum runs over all paths through v. For clique size two, the witnesses are edges, so the update becomes the standard core fixed-point test over neighbors [22, 32]. For larger clique sizes, the counted items are clique witnesses, not distinct neighbor vertices. The next lemma counts $\sum_{\mathcal{P} \ni v} W_v(\mathcal{P}, h, k)$ from CPI paths without enumerating those witnesses.

Lemma 5.1 (Per-path Witness Count)**.** *Fix a vertex v, a current vector h, and a threshold $k \geq 1$. For a path $\mathcal{P}$ with $v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})$, let $H_k(\mathcal{P})$ and $Q_k(\mathcal{P})$ be the number of vertices in $V_h(\mathcal{P}) \setminus \{v\}$ and $V_p(\mathcal{P}) \setminus \{v\}$, respectively, whose current h-value is at least k. The number $W_v(\mathcal{P}, h, k)$ of encoded s-cliques through v in which every other vertex has h-value at least k is:*

(1) if $H_k(\mathcal{P}) < \lvert V_h(\mathcal{P}) \setminus \{v\} \rvert$, then $W_v(\mathcal{P}, h, k) = 0$;

(2) if $v \in V_h(\mathcal{P})$, then $W_v(\mathcal{P}, h, k) = \binom{Q_k(\mathcal{P})}{\eta_{\mathcal{P}}}$;

(3) if $v \in V_p(\mathcal{P})$, then $W_v(\mathcal{P}, h, k) = \binom{Q_k(\mathcal{P})}{\eta_{\mathcal{P}} - 1}$.

Summing over the paths through v gives the total threshold count $\sum_{\mathcal{P} \ni v} W_v(\mathcal{P}, h, k)$, and $\Phi_h(v)$ is the largest k for which this sum is at least k.

Proof. Each encoded s-clique through v has the form $V_h(\mathcal{P}) \cup P'$ with $P' \subseteq V_p(\mathcal{P})$ and $\lvert P' \rvert = \eta_{\mathcal{P}}$. If any other hold vertex fails threshold k, no such clique qualifies; otherwise, $v \in V_h(\mathcal{P})$ leaves all $\eta_{\mathcal{P}}$ pivot slots to be filled from the $Q_k(\mathcal{P})$ passing pivots and gives $\binom{Q_k(\mathcal{P})}{\eta_{\mathcal{P}}}$ qualifying cliques, while $v \in V_p(\mathcal{P})$ leaves $\eta_{\mathcal{P}} - 1$ slots to fill and gives $\binom{Q_k(\mathcal{P})}{\eta_{\mathcal{P}} - 1}$. Summing over the paths through v gives the stated total. □ □

Algorithm 2: VertexSupport(v, h) — one-vertex update from static clique witnesses.

Input: Vertex $v \in V$; value array h with $h[u]$ the running estimate of $\kappa_s(u)$ for each $u \in V$; CPI $\mathcal{P}$.
Output: The threshold-support value $\Phi_h(v)$.

```
1 C[0, ..., h[v]] ← 0 ;                          // threshold witness counts
2 foreach path P ∈ P through v do
3     for k ← 1 to h[v] do
4         C[k] ← C[k] + W_v(P, h, k);                          // Lemma 5.1
5 for k ← h[v] to 1 do
6     if C[k] ≥ k then return k;
7 return 0
```

VertexSupport(v, h) computes this update for one vertex. During one call, the algorithm builds only local threshold state for v. After any prefix of the paths through v, that local state contains exactly the witnesses contributed by the scanned paths. Scanning the next path adds the binomial counts from Lemma 5.1, and the largest threshold with enough witnesses is the updated value. Because the call reads the static CPI but never mutates a path, the same witness rule can be used by both solvers below.

Algorithm 2 zeroes the local threshold-count array up to $h[v]$ (Line 1), accumulates the per-threshold witness count $W_v(\mathcal{P}, h, k)$ into $C[k]$ for every path through v using the closed form of Lemma 5.1 (Lines 2-4), and returns the largest k at which $C[k] \geq k$ by scanning C from $h[v]$ downward (Lines 5-7); otherwise it returns 0. A direct implementation does not enumerate clique witnesses: it evaluates the same threshold counts from the values stored on each scanned path, using sorting or buckets, so one call costs $O(\deg_\Sigma(v) \log \deg_\Sigma(v))$. The call reads the static CPI and the vector h, and it mutates no path.

5.2 Direct Static-Path Iteration

SPIN is the most literal solver of the threshold-support equation: it scans all vertices, computes a next value for each vertex from the previous value vector via VertexSupport, and repeats until a full scan makes no change. The only mutable data are the current and next value vectors; the current vector starts at the initial s-clique support, so every entry is an upper bound on the final core value, and each pass can only lower entries. A fixed point means every

Algorithm 3: SPIN — direct static-path fixed-point iteration.

Input: Graph G, clique size s, CPI $\mathcal{P}$.
Output: $\kappa_s(v)$ for every $v \in V$.

1 **foreach** $v \in V$ **do**
2 $h[v] \leftarrow \deg^{(s)}(v)$; // closed form from Lemma 2.6
3 changed $\leftarrow$ true;
4 **while** changed **do**
5 changed $\leftarrow$ false; $h_{\text{new}} \leftarrow h$;
6 **foreach** $v \in V$ in a fixed order **do**
7 $h_{\text{new}}[v] \leftarrow \text{VertexSupport}(v, h)$;
8 **if** $h_{\text{new}}[v] \neq h[v]$ **then** changed $\leftarrow$ true;
9 $h \leftarrow h_{\text{new}}$;
10 **foreach** $v \in V$ **do**
11 $\kappa_s(v) \leftarrow h[v]$;
12 **return** κ_s

vertex value is consistent with the witnesses whose other vertices have high enough current value, which makes SPIN the reference formulation for the static-CPI computation even though repeated full scans can be expensive.

Algorithm 3 initialises $h[v]$ to the initial s-clique support $\deg^{(s)}(v)$ for every vertex via the closed form of Lemma 2.6 (Lines 1-2). The main loop (Lines 3-8) iterates until no entry changes in a full pass: it sets a *changed* flag to false (Line 5), copies the current vector into h_{new} (Line 5), then calls VertexSupport(v, h) for every vertex in a fixed order and stores the result into $h_{\text{new}}[v]$ (Lines 6-7); if any update lowered an entry the flag is set, otherwise the pass exits. The swap $h \leftarrow h_{\text{new}}$ at the end of the pass (Line 8) avoids order dependence inside the pass. The final core values are read off the converged vector (Lines 9-10) and returned (Line 11). A pass with no changed entry is the certificate that the fixed point has been reached.

Example 5.2 (Static-path update on the running example). Initialise $h[v] \leftarrow \deg^{(s)}(v)$ using Lemma 2.6; on the running example $h[v_4] = 6$ (contributions 2+1+2 from $\mathcal{P}_3, \mathcal{P}_4, \mathcal{P}_5$ as pivot and 1 from $\mathcal{P}_6$ as hold; see Table 1). The first call VertexSupport(v_4, h) scans the four paths through v_4 and counts threshold witnesses via Lemma 5.1. At threshold $k = 3$, every other vertex on each of the four paths has $h \geq 3$, so the witness sums are 2+1+2+1 = 6; the largest k with $\sum_{\mathcal{P} \ni v_4} W_{v_4}(\mathcal{P}, h, k) \geq k$ is $\Phi_h(v_4) = 3$, the final $\kappa_3(v_4)$. SPIN reaches the fixed point in a small number of full passes, but each pass scans every vertex regardless of whether its value can still change; this is the redundancy that SPIN* eliminates in Section 5.3.

Lemma 5.3 (Fixed Point Correctness). *Initialised at $h^{(0)}[v] = \deg^{(s)}(v)$, Algorithm 3 converges in at most $1 + n\kappa_{\max}$ passes to $h[v] = \kappa_s(v)$ for every $v \in V$, and the sequence $h^{(t)}[v]$ is monotonically non-increasing.*

Proof. By the operator definition, $h^{(t+1)}[v] \geq k$ iff v has at least k incident s-cliques whose other vertices all satisfy $h^{(t)} \geq k$, so $\deg^{(s)}(v)$ is the largest possible value and the sequence is monotone non-increasing on the non-negative integers; the total drop across all vertices is bounded by $n\kappa_{\max}$, giving $T \leq 1 + n\kappa_{\max}$ passes to a fixed point $h^\star$. At $h^\star$, let $S_k = \{v \mid h^\star[v] \geq k\}$; every $v \in S_k$ has at least k incident s-cliques entirely inside S_k, and each such clique lies in v's s-connected component $C \subseteq S_k$ (because the clique itself is the connecting witness), so C satisfies Conditions (i)–(iii) of Definition 2.3 and is a k-$(1, s)$-nucleus, giving $\kappa_s(v) \geq k$. Conversely, if v lies in a k-$(1, s)$-nucleus C', induction on t shows that the threshold-support equation preserves $h^{(t)}[u] \geq k$ for every $u \in C'$, so $h^\star[v] \geq k$. Hence $h^\star[v] = \kappa_s(v)$. □ □

Theorem 5.4 (SPIN Complexity). *Algorithm 3 computes κ_s in $O(T \cdot \Sigma \cdot \log d_{\max})$ work, where $T \leq 1 + n\kappa_{\max}$ is the number of passes.*

Proof. Each pass calls VertexSupport(v, h) for every vertex v. The cost of one call is $O(\deg_\Sigma(v) \log \deg_\Sigma(v))$. Summing over vertices gives $O(\Sigma \log d_{\max})$ work per pass. Lemma 5.3 bounds the number of passes. □ □

5.3 Eliminating Redundant Sweeps

SPIN is useful because it exposes the equations, but its full passes recompute many vertices whose values can no longer change. Once a vertex reaches its final peel level, later events cannot increase it. SPIN* removes this redundancy by using the same nondecreasing peel order as classical core decomposition. The algorithm keeps the same value semantics as SPIN, but it updates them with local path events instead of global passes. For each path, it stores whether the path is still live, how many pivot vertices have not been finalized, and the fixed pivot requirement. It also keeps a finalized set and a bucket queue keyed by the current value. When one minimum vertex is popped, only paths containing that vertex can change. Theorem 4.1 says how each such path changes, and Lemma 4.2 gives the exact support loss for the unfinalized path-mates. Thus the invariant is local: unfinalized vertex values reflect the current live path-counter contributions, clipped below by the current peel level.

SPIN* is an optimization of SPIN, not a different decomposition objective. It keeps the same witness equation but changes the schedule. Instead of scanning every vertex after every pass, it updates only the vertices whose path contributions changed. Algorithm 4 outputs the core values and emits a sorted deletion log as a byproduct. Its running time depends on U, the number of incremental support-update events that actually occur, rather than on the number of full scans.

Algorithm 4 first initialises every path's pivot count $p_{\mathcal{P}}$, pivot requirement $\eta_{\mathcal{P}}$, and liveness bit (Lines 1-2), then sets each vertex's working value $h[v]$ to its initial s-clique support and inserts v into the corresponding bucket (Lines 3-4). The main loop (Lines 6-15) advances the peel level k to the smallest non-empty bucket and pops the next vertex v (Line 7), finalises $\kappa_s(v) \leftarrow k$ (Line 8), and walks every live path that contains v (Lines 10-13): if v is a hold, the path dies (Line 11); if v is a pivot, the path's pivot counter decrements and the path also dies once $p_{\mathcal{P}} < \eta_{\mathcal{P}}$ (Lines 12-13). Affected path-mates that have not been finalised are recomputed via Lemma 4.2 (Line 14) and moved to a lower bucket if their value dropped (Line 15). The pseudocode writes the affected update as VertexSupport to keep the fixed-point semantics visible; the implementation evaluates the changed contribution with the binomial loss formula rather than recomputing from scratch, so the total work scales with the number of incremental update events U rather than with the number of full scans. Buckets maintain the minimum-h vertex because values only decrease and are bounded by $\kappa_{\max}$ [6].

Lemma 5.5 (SPIN* Equivalence). *Algorithm 4 returns the same κ_s values as Algorithm 3.*

Algorithm 4: SPIN* — core values via single-pass bucket-queue peeling.

Input: Graph G, clique size s, CPI $\mathcal{P}$.
Output: $\kappa_s(v)$ for every $v \in V$.

foreach *path* $\mathcal{P} \in \mathcal{P}$ **do**
  $p_{\mathcal{P}} \leftarrow |V_p(\mathcal{P})|$; $\eta_{\mathcal{P}} \leftarrow s - |V_h(\mathcal{P})|$; alive$_{\mathcal{P}} \leftarrow (p_{\mathcal{P}} \geq \eta_{\mathcal{P}})$;
foreach $v \in V$ **do**
  $h[v] \leftarrow \deg^{(s)}(v)$; insert v into bucket $B[h[v]]$;
$k \leftarrow 0$; $V_{\text{fin}} \leftarrow \emptyset$;
while $|V_{\text{fin}}| < n$ **do**
  advance k until $B[k] \neq \emptyset$; pop v from $B[k]$;
  $\kappa_s(v) \leftarrow k$; append (v, k) to the deletion log; $V_{\text{fin}} \leftarrow V_{\text{fin}} \cup \{v\}$;
  Aff $\leftarrow \emptyset$;
  foreach $\mathcal{P}$ *with* alive$_{\mathcal{P}}$ *and* $v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ **do**
    Aff $\leftarrow$ Aff $\cup$ $((V_h(\mathcal{P}) \cup V_p(\mathcal{P})) \setminus \{v\})$;
    if $v \in V_h(\mathcal{P})$ **then**
      alive$_{\mathcal{P}} \leftarrow$ false; `// path death`
    else
      $p_{\mathcal{P}} \leftarrow p_{\mathcal{P}} - 1$;
      if $p_{\mathcal{P}} < \eta_{\mathcal{P}}$ **then** alive$_{\mathcal{P}} \leftarrow$ false;
  foreach $u \in$ Aff $\setminus V_{\text{fin}}$ **do**
    $y \leftarrow \max\{k, \text{VertexSupport}(u, h)\}$; `// implemented incrementally with Lemma 4.2`
    if $y < h[u]$ **then** move u from $B[h[u]]$ to $B[y]$; $h[u] \leftarrow y$;
return κ_s

Proof. Both algorithms apply the same per-vertex threshold-support operator Φ: SPIN uses full passes, while SPIN* (by Theorem 4.1 and Lemma 4.2) restricts recomputes to path-mates of the popped vertex. Initial values $h[u] = \deg^{(s)}(u)$ are an upper bound on $\kappa_s(u)$ and Φ never increases an entry, so $h \geq \kappa_s$ holds throughout. When v is popped at level $k = h[v]$, every unfinalised u has $h[u] \geq k$ and at least k surviving s-cliques inside the unfinalised set U; each such clique lies in u's s-connected component of U, which is therefore a k-$(1, s)$-nucleus, so $\kappa_s(u) \geq k$, and combined with $h[u] \geq \kappa_s(u)$ this gives $\kappa_s(v) = k = h[v]$. By Lemma 5.3, κ_s is also the fixed point reached by SPIN. □ □

Theorem 5.6 (SPIN* Complexity). *Algorithm 4 runs in $O(\Sigma + U + \kappa_{\max} + n)$ time, where U is the total number of incremental support-refresh events over all affected path-mates.*

Proof. Initialization of vertex values and path counters costs $O(n + \Sigma)$. Each vertex–path incidence is visited when its vertex is finalized, so path-event processing costs $O(\Sigma)$. Each affected incremental refresh contributes one unit to U. Bucket moves and bucket pops are $O(1)$ amortized. The bucket cursor advances at most $\kappa_{\max}$ times. The total time is therefore $O(\Sigma + U + \kappa_{\max} + n)$. □ □

The mutable baseline pays heap and residual-index maintenance costs on the same logical peel events. The experiments in Section 8 measure the impact of replacing those operations with static path counters.

6 Parallel CPI Construction

Once SPIN* makes peeling counter-only, CPI construction becomes the load-bearing cost of the pipeline: the phase-breakdown tables of Section 8 report `web-it-2004` at $s = 3$ peeling in 351 ms against $\approx$ 127 s of construction, and `web-Stanford` at $s = 60$ peeling in 1.4 ms against $\approx$ 5.8 s of construction. We parallelise the construction phase by exploiting two properties: the seed loop partitions emitted

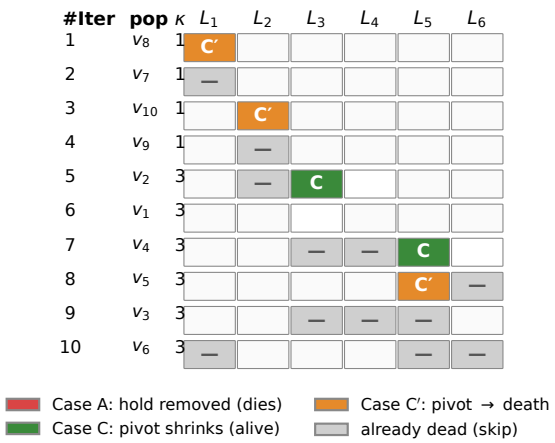

Figure 2: Incremental peel trace on the running example.

Algorithm 5: ParaBuild — parallel CPI construction via per-thread path buffers.

Input: Graph G in CSR; clique size s; thread count T.
Output: Static CPI $\mathcal{P}$ in shared CSR.

compute degeneracy ordering σ on G;
launch T threads and assign each thread t a contiguous chunk σ_t of seed vertices;
foreach *thread* $t \in \{1, \ldots, T\}$ in parallel **do**
  $A_t \leftarrow \emptyset$; `// thread-local path arena`
  $b_t \leftarrow$ generation-stamped bitmap over V;
  foreach *seed vertex* $v \in \sigma_t$ **do**
    increment generation of b_t;
    $N^+ \leftarrow \{u \in N_G(v) \mid \sigma(u) > \sigma(v)\}$;
    BkEmit(v, N^+, A_t, b_t) ; `// pivoted BK on` $G[N^+]$`, with` v `as a hold`
$\mathcal{P} \leftarrow$ deterministic concat-and-prefix-sum merge of $\{A_t\}_{t=1}^{T}$;
build incidence CSR over $\mathcal{P}$ in parallel;
return $\mathcal{P}$

s-clique families along the degeneracy order, and the static output contract makes the merge a pointer-level concatenation.

Why the seed loop can be parallel. The CPI builder follows the standard degeneracy-ordered Bron–Kerbosch search with Tomita pivoting [4, 10, 16, 45]. Each s-clique has one owner: the smallest vertex of the clique in the degeneracy order. Therefore, the outer loop over seed vertices partitions the emitted s-clique families. No two seeds own the same s-clique.

Static CPI also changes what construction must produce. The mutable baseline builds residual tree state because later peeling may edit that state. Our $r = 1$ pipeline never edits the stored hold or pivot sets. The builder therefore only has to emit a path list and vertex–path incidence arrays. This output can be accumulated in thread-local arenas and merged after all seeds finish.

Thread-local construction. ParaBuild takes G, s, and a thread count. Each worker owns a contiguous seed range, a local path arena, and a generation-stamped membership bitmap. This ownership is enough because seed ranges partition the emitted clique families. While a worker processes its seeds, it writes only to its own arena. After all workers finish, a deterministic merge concatenates the arenas and builds the shared incidence CSR. The expensive work is still the Bron–Kerbosch search under the seeds; the merge is linear in the emitted incidences.

Algorithm 5 computes the degeneracy ordering σ on G (Line 1) and partitions the seed range into T contiguous chunks (Line 2). Each thread allocates a private path arena and a generation-stamped membership bitmap (Lines 4-5), then walks its seed range (Lines 6-9): for every seed v it increments the bitmap generation (Line 7), restricts the neighbourhood to vertices later in σ (Line 8), and runs the pivot-aware enumeration BkEmit streaming each emitted path into the thread-local arena (Line 9). After all threads finish, a deterministic concat-and-prefix-sum merge concatenates the arenas (Line 10) and builds the shared incidence CSR in parallel (Line 11); the final static CPI is returned (Line 12). BkEmit is the same pivot-aware enumeration used by the CPI construction lineage [4, 24]; the difference is that it streams each emitted path into A_t instead of storing a mutable recursion tree for later residual updates.

Correctness. ParaBuild is a parallel schedule of the same seed-owned enumeration. Seed ownership gives disjoint emitted s-clique families. The deterministic merge changes only the storage order, not the represented path families. The resulting path set satisfies Definition 2.5. Therefore, the support formula of Lemma 2.6 applies unchanged.

Scope. The design parallelizes construction because construction is the phase exposed by the static-CPI pipeline. It does not claim a lower bound showing that SPIN* cannot be parallelized. It also does not claim ideal scaling. The scaling evidence is reported in Figure 6.

7 Core-Value Hierarchy Construction

Because the static-CPI invariant of Section 4 keeps every leaf intact through peeling, the hierarchy is no longer a separate decomposition problem: every CPI leaf encodes the same family of s-cliques both before and after the peel, so the elder-rule join tree of Definition 2.3 can be emitted by a single descending union–find scan over the preserved leaves, in $O(\Sigma \log \Sigma)$ time, without re-enumerating a single clique. Algorithms that mutate the CPI during peeling cannot reuse it for this purpose: they must either re-run s-clique enumeration at each level or maintain one parallel disjoint-set forest per core level during peeling, which blows up memory by an order of magnitude (Section 8, Figure 9). For any level k, the super-level set is $V_k = \{v \mid \kappa_s(v) \geq k\}$ and Definition 2.3 measures connectivity inside V_k through s-clique witnesses contained in V_k; BuildHier (Algorithm 6) materialises that filtration from the preserved leaves.

Leaves as connectors. A CPI leaf P encodes the family of s-cliques $V_h(P) \cup X$ for every η_P-subset $X \subseteq V_p(P)$ (Definition 2.5). Define the *birth level* of P as

$$k_P = \min_{v \in V_h(P) \cup V_p(P)} \kappa_s(v).$$

At level $k \leq k_P$ every leaf vertex has core value at least k, so every encoded s-clique is contained in V_k. Any two vertices of the leaf either lie inside one such clique (when $\eta_P \geq 2$) or are linked by a length-two path through a vertex of $V_h(P)$ (when $\eta_P = 1$ and $V_h(P)$ is non-empty); for $V_h(P) = \emptyset$ and $\eta_P \geq 2$, any pair appears together in some s-subset. In every case the leaf vertices form a single s-connected component the moment P becomes active. This lets BuildHier treat each leaf as a single hyperedge in a union–find structure: the leaf node is unified with every vertex it contains. Two vertices then end up in the same component if and only if they are s-connected through active CPI cliques at the current level.

Algorithm 6: BuildHier — elder-rule join tree atop SPIN*.

Input: Graph G, clique size s, CPI $\mathcal{P}$.
Output: Elder-rule join tree T of the s-clique-connected super-level filtration.

1 $\kappa_s \leftarrow$ SPIN$^\star(G, s, \mathcal{P})$; // Algorithm 4
2 **foreach** *leaf* $P \in \mathcal{P}$ **do**
3 $k_P \leftarrow \min\{\kappa_s(v) \mid v \in V_h(P) \cup V_p(P)\}$;
4 sort $\mathcal{P}$ by nonincreasing k_P (ties broken by leaf id);
5 initialise union-find U over $V \cup \{P \mid P \in \mathcal{P}\}$; birth$[r] \leftarrow +\infty$ for every root r;
6 $T \leftarrow$ empty forest;
7 **foreach** *leaf* P *in sorted order* **do**
8 $R \leftarrow \{\textsc{Find}(x) \mid x \in V_h(P) \cup V_p(P) \cup \{P\}\}$;
9 **foreach** *root* $r \in R$ **do**
10 **if** *birth*$[r] = +\infty$ **then**
11 emit a hierarchy node n_r with $k_{\text{birth}} \leftarrow k_P$; birth$[r] \leftarrow k_P$;
12 **if** $|R| \geq 2$ **then**
13 sort R descending by $(\text{birth}[r], |r|)$; let $r^\star$ be the elder ; // elder rule
14 **foreach** $r \in R \setminus \{r^\star\}$ **do**
15 attach n_r as child of $n_{r^\star}$ at level k_P in T ; // younger dies
16 Union all of R into a single component;
17 **return** T

Reverse scan over leaves. BuildHier processes leaves in nonincreasing k_P, as the level descends through the filtration. A union–find structure spans both vertices and leaf identifiers; processing leaf P at its birth level fires unions $\textsc{Union}(P, v)$ for every $v \in V_h(P) \cup V_p(P)$. A birth level is recorded for each component root, so a merge applies the elder rule. Vertices not appearing in any leaf with $k_P \geq 1$ have $\kappa_s(v) = 0$ and are omitted from the hierarchy.

Algorithm 6 first invokes SPIN* to compute the core values (Line 1), then assigns each leaf its birth level k_P as the minimum core value among its hold and pivot vertices (Lines 2-3). Leaves are sorted by nonincreasing k_P (Line 4) so they fire in descending peel order, and the union–find structure is initialised over both vertices and leaf identifiers with all roots born at $+\infty$ (Line 5). For each leaf P in sorted order (Lines 7-13), the algorithm collects the current root set R of every vertex in P together with the leaf itself (Line 8), emits a fresh hierarchy node for any root that has not been born yet (Lines 9-11), applies the elder rule when $|R| \geq 2$ by selecting the earliest-born root $r^\star$ and attaching the others as its children at level k_P (Lines 12-14), and finally unifies all roots into a single component (Line 15). The returned forest T is the elder-rule join tree.

Correctness. The descending scan visits each leaf at exactly the level it first contributes s-cliques contained in the current super-level set. At that level, every vertex it contains lies in the super-level set, so the leaf witnesses pairwise s-connectivity for all of them through a common element of $V_h(P)$ (or directly within an s-subset when $V_h(P) = \emptyset$). Conversely, every active s-clique at level k belongs to some leaf with $k_P \geq k$, and that leaf has already fired before the scan reaches level k; so every s-clique-connected merge of Definition 2.3 is performed. When two distinct components meet at a leaf, the elder rule preserves the component with earlier birth and attaches the younger one in the join tree [15].

Complexity. Computing every k_P and sorting leaves costs $O(\Sigma + L \log L)$, where $L = |\mathcal{P}|$. Each leaf fires at most $|V_h(P)| + |V_p(P)|$ union-find operations, summing to $O(\Sigma\, \alpha(n + L))$ over all leaves.

Total time $O(\Sigma \log \Sigma)$, dominated by the sort, with no clique re-enumeration: BuildHier reads each CPI leaf at most twice (once to compute k_P, once to fire unions).

Memory. Beyond the union-find of size $n + L$, the routine needs a list of vertices per leaf to fire unions. SPIN*'s vertex-to-path inverse CSR (Section 5) supports a single bucket-CSR pass that materialises the per-leaf lists in $O(\Sigma)$ temporary memory; this scratch is freed on return. The static CPI budget of Section 8 therefore holds throughout peeling proper and is only briefly elevated during the hierarchy emit.

Consequence. The clique-coherent core-value hierarchy is not a separate decomposition problem in this pipeline. It is a byproduct of the static CPI surviving peeling: leaves act as hyperedges of an implicit s-clique witness graph, and a single descending pass with union–find produces the join tree. Methods that mutate the CPI during peeling cannot reuse it for hierarchy emit and need a second clique-enumeration pass [40].

Section 8 (Exp-7) compares BuildHier against the only feasible baseline at r=1 — CND's general (r, s) level-DSU hierarchy specialised to r=1 — and confirms that BuildHier uses up to 43× less memory (gmean 13× over 24 matched cells) and up to 35× less hierarchy-emission wall time, with the gap growing at larger s where the witness mass dominates.

8 Experiments

Exactness comes from the invariants and fixed-point equivalence proved earlier; the experiments measure cost. **Exp-1** measures end-to-end time and peak memory for SPIN*, the practical static-CPI pipeline. **Exp-2** separates construction from peeling, because counter-only peeling shifts rather than removes all cost. **Exp-3** tests how the static construction contract scales with thread count. **Exp-4** measures SPIN as the direct fixed-point baseline. **Exp-5** pushes the pipeline to a billion-edge graph (`com-friendster`, 1.8B edges). **Exp-6** sweeps a vertex-induced subsampling ratio to test how runtime and memory scale with input size. **Exp-7** compares hierarchy emission against CND's level-DSU baseline.

Hardware and software. All experiments run on a dual-socket Intel Xeon Gold 6248*R* machine with 24 cores and 503 GB DDR4 memory. The operating system is Ubuntu 22.04. The code[1] is compiled with Clang 18.1 using `-O3 -march=native`. It is linked against TBB 2021.10, Boost 1.83, and `robin_hood::unordered_flat_map`. Single-threaded measurements use one core. Parallel-construction measurements use the thread counts reported in Figure 6. Time and memory measurements are averaged or summarized over repeated runs as stated in each experiment.

Datasets. Table 2 lists the benchmark graphs. The set covers collaboration, e-commerce, social, and web graphs. The largest graph is `com-friendster` with 1.8B edges. The benchmark sweep on the ten smaller graphs yields 307 matched cells where both SPIN* and CND complete successfully, and the headline speedup numbers are reported over the matched subset; `com-friendster` is run standalone (Section 8) because CND cannot complete on it.

Algorithms and methodology. CND [4] is the mutable-CPI implementation described in Section 3, restricted to $r = 1$ for this comparison; SPIN* is the static-CPI pipeline of this paper; SPIN is the direct fixed-point reference of Section 5.2. All three solvers consume the same input edge list and run on the same hardware; the comparison therefore isolates the cost of CPI mutability against static-state peeling. For each matched $(graph, s)$ pair, we record the end-to-end time (load, degeneracy order, CPI construction, peel, output) and the program's memory usage.

[1] We will release the code, bench harness, and result CSVs under an anonymised public archive at submission and de-anonymise the repository on acceptance.

Table 2: Datasets.

Graph	$\lvert V\rvert$	$\lvert E\rvert$	avg. deg.
`com-amazon` [49]	334,863	925,872	5.5
`com-dblp` [49]	317,080	1,049,866	6.6
`twitter` [49]	81,306	1,342,296	33.0
`web-Stanford` [49]	281,903	1,992,636	14.1
`com-youtube` [49]	1,134,890	2,987,624	5.3
`web-Google` [49]	875,713	4,322,051	9.9
`wiki-Talk` [49]	2,394,385	4,659,565	3.9
`web-it-2004` [49]	509,338	7,178,413	28.2
`soc-pokec` [49]	1,632,803	22,301,964	27.3
`com-orkut` [49]	3,072,441	117,185,083	76.3
`com-friendster` [49]	65,608,366	1,806,067,135	55.1

Table 3: Per-incidence memory budget at $\Sigma \gg n$, where $\Sigma = \sum_{P \in \mathcal{P}} \big(|V_h(P)| + |V_p(P)|\big)$ is the total vertex–path incidence count (Section 2).

Component	**Baseline**	SPIN*
BK recursion tree T	8Σ	0
Path-to-vertex storage (V_h, V_p)	0	4Σ
Per-path vertex set H_P	32Σ	—
Reverse vertex-to-path map	48Σ	—
Vertex-to-path incidence	—	4Σ
Support array	$4n$	$4n$
Heap / bucket array	$24n$	$16n + 8\kappa_{\max}$
Analytical Σ-coefficient	$\approx 88\Sigma$	$\approx 8\Sigma$
Measured: `com-youtube`, s=5	642 MB	387 MB
Measured: `web-it-2004`, s=3	556 MB	270 MB
Measured: `web-Stanford`, s=10	261 MB	147 MB

Exp-1: end-to-end time and memory. Across the ten graphs (Figures 3–4), SPIN* achieves a peel-time speedup over CND of gmean 13.50× / max 44.76×, and a memory-usage ratio (CND/SPIN*) of gmean 1.55× / max 3.92×. The end-to-end speedup is smaller (gmean 1.10×) because the shared CPI construction phase dominates total runtime once peeling is cheap; this is the bottleneck shift that motivates ParaBuild in Section 6.

The memory improvement is explained by the incidence-level budget in Table 3. The mutable baseline stores path hash sets and reverse hash maps. SPIN* stores flat path-to-vertex and vertex-to-path arrays plus path counters. The analytical per-incidence gap is larger than the observed memory gap because graph adjacency, allocator overhead, and per-vertex structures do not scale with Σ.

Exp-2: phase breakdown. Counter-only peeling is cheap; we measure how much of the remaining cost lies in CPI construction. The phase study sweeps $s \in \{3, 5, 7, 9, 11, 13, 15\}$ on six graphs (Figure 5, smallest to largest by edge count). On every graph, the orange peel slice on top of each bar shrinks as s grows: on the medium and large graphs peel falls below 1% of total time by s=9 and stays below 0.1% on `web-it-2004` throughout the sweep. The bottleneck has shifted onto construction, which is why ParaBuild is

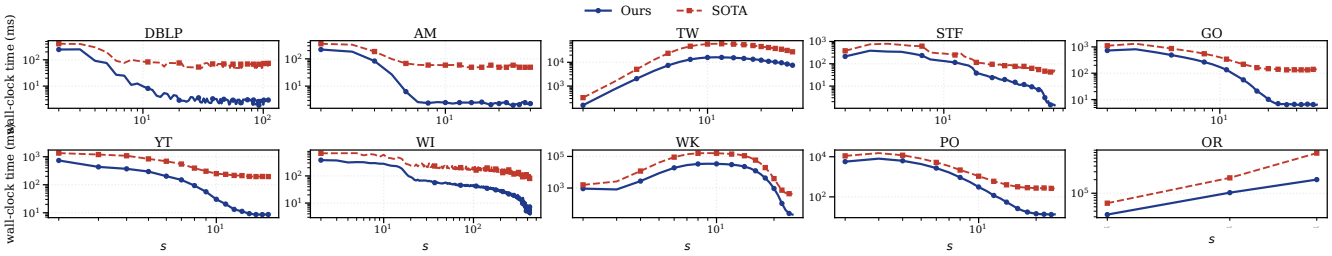

Figure 3: End-to-end time across ten graphs.

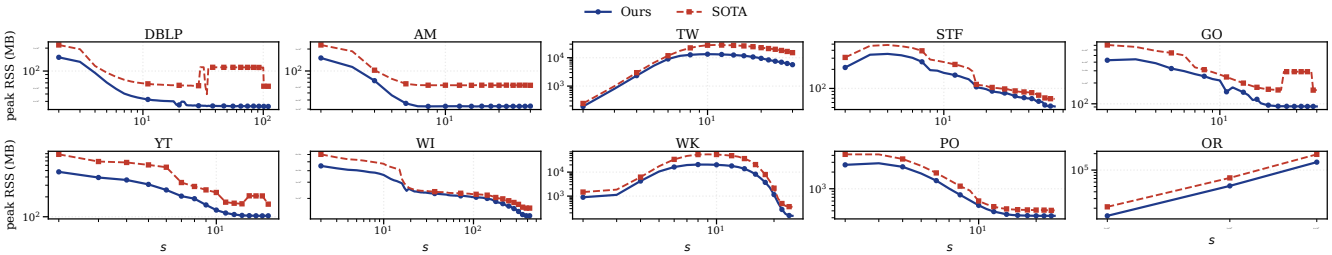

Figure 4: Memory usage across ten graphs.

part of the algorithmic pipeline rather than a detached engineering detail.

Exp-3: parallel scaling of CPI construction. ParaBuild exposes seed-level parallelism because each seed can stream its emitted paths into thread-local storage. Figure 6 reports build time on six graphs as the thread count grows from T=1 to T=64 (median of three runs per cell), with one panel per graph showing the representative s=3 curve; each curve ends at its optimum thread count, and graphs with T=1 baseline below 500 ms are omitted as uninformative. On large graphs the optimum lies at T=24–48; on smaller graphs it is around T=16. The headline single-cell speedup is 23.4× at T=24 on `web-it-2004` s=3 (92.3 s → 3.95 s).

Exp-4: SPIN⋆ vs. SPIN on the same fixed point. SPIN is the literal single-thread solver for the same equations that SPIN⋆ realizes by deletion events. The two algorithms compute the same core values (Lemma 5.5), so the comparison isolates the effect of replacing repeated full-graph passes with incremental updates. Both solvers read paths from the same static index; the SPIN reference therefore reflects the algorithmic cost of repeated full-pass fixed-point iteration, not a difference in how the index is stored. Across the 272 matched $(graph, s)$ cells, SPIN⋆ delivers a 30.4× geometric-mean peel-time speedup over SPIN, with maximum 2,369× on `web-it-2004` at $s = 53$, where SPIN repeatedly sweeps every vertex while SPIN⋆ finishes after a few path-event updates. On the easiest cells SPIN⋆ can be slightly slower (minimum ratio 0.62×) because the bucket-queue setup and live-path bookkeeping add fixed overhead that SPIN's tight inner loop avoids when one or two passes already converge. On the largest inputs SPIN hits the 1h cap (`com-orkut` for $s \geq 3$, `wiki-Talk` for the single cell $s = 10$); SPIN⋆ finishes the same configurations in seconds to minutes.

SPIN's memory profile is essentially identical to SPIN⋆'s on every paired cell, because both share the same static index: the geometric-mean memory ratio (SPIN/SPIN⋆) is 0.97 over the 291 memory-paired cells (more than the 272 time-paired cells because some non-finishing SPIN runs still recorded a peak-RSS value), and SPIN is slightly smaller than SPIN⋆ on 83.8% of them. The reason is structural: SPIN⋆'s extra state beyond the shared CPI consists of a per-path counter cache (∼ 13 bytes per leaf) and a peel-order bucket queue (∼ 17 bytes per vertex), which SPIN does not maintain because it scans every vertex each round. The sub-percent memory advantage of SPIN is therefore the price SPIN⋆ pays to avoid repeated full-graph passes; the trade is the textbook "space for time" choice, with the time term dominating by orders of magnitude.

The paper carries both algorithms because they serve different roles: SPIN is the direct semantic formulation, SPIN⋆ is its event-driven realisation.

Exp-5: scaling to a billion-edge graph. The ten-graph benchmark stops at `com-orkut` (117M edges); to push the pipeline an order of magnitude further, we run SPIN⋆ with T=24 on `com-friendster` (65M vertices, 1.806B edges, the largest SNAP graph available). CND completes only at s=2, using 400 GB of peak memory and ≈ 30 minutes of peel against SPIN⋆'s 160 GB and 344 s (Figure 7: ≈ 5.2× peel speedup, 2.5× less memory on the matched cell); for $s \geq 3$ it exceeds the 503 GB budget during construction and is OOM-killed. SPIN⋆ decomposes the graph for a wide range of s: at s=6 it indexes $\Sigma \approx 2.26 \times 10^{10}$ encoded s-clique witnesses and peels that entire witness mass in 651 s; at s=11 it indexes $\Sigma \approx 1.68 \times 10^{10}$ witnesses and finishes peeling in 167 s. Construction dominates total runtime (5–7 thousand seconds of build vs. 167–676 seconds of peel), confirming that the bottleneck shift carries over to the billion-edge regime; the parallel builder of Section 6 is what makes the construction phase tractable on a 1.8B-edge input.

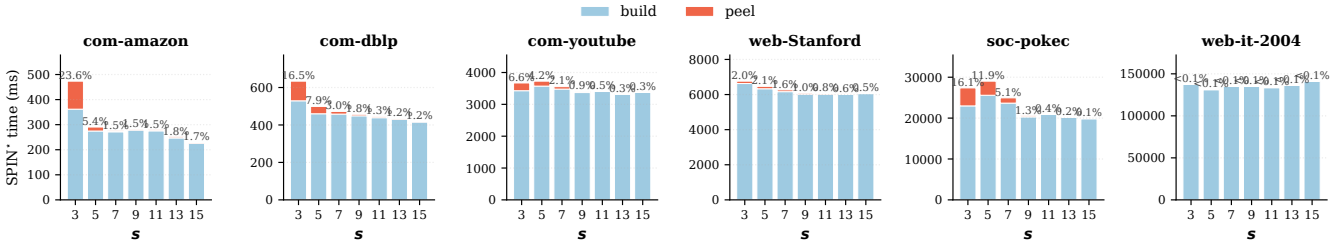

Figure 5: SPIN* **phase breakdown on six graphs: build (blue, bottom) vs peel (orange, top). Linear scale; the peel share is printed above each bar.**

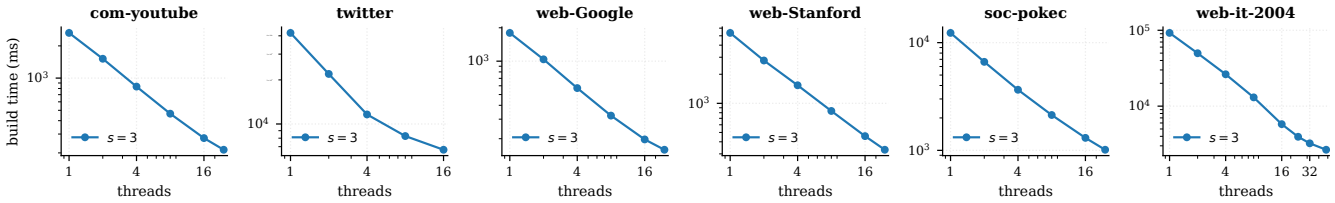

Figure 6: ParaBuild **build time vs. thread count; one curve per representative** s**, ending at its optimum** T**.**

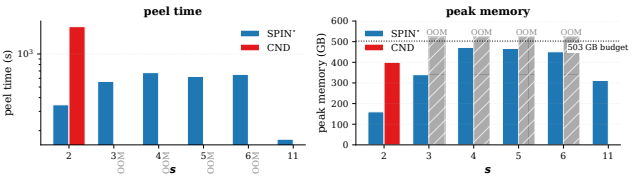

Figure 7: `com-friendster` **(1.806B edges): peel time (left) and peak memory (right).** CND **completes only at** s=2**; for** $s \geq 3$ **it exceeds the** 503 **GB budget.**

Exp-6: scaling with input size. The benchmark cells fix a graph and sweep s; they do not test how runtime and memory respond to growing the input itself. To probe scaling, we shrink each graph by vertex-induced subsampling: for each ratio $\rho \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ we keep $\lceil \rho\, n \rceil$ vertices selected uniformly at random with a fixed seed and take the induced subgraph (only edges with both endpoints kept). The same seed makes the subsamples nested ($20\% \subset 40\% \subset \cdots \subset 100\%$), so runtime and memory are monotone in ρ by construction. We pick vertex-induced rather than random-edge subsampling because SPIN*'s work scales with the encoded s-clique count Σ; removing edges uniformly drops Σ super-linearly because each missing edge can destroy many cliques, producing jagged curves, while a vertex-induced subsample preserves clique structure inside the kept vertex set. We re-run SPIN* on six graphs spanning over two orders of magnitude in edge count (`com-dblp`, `com-youtube`, `web-Stanford`, `web-Google`, `soc-pokec`, `com-orkut`) for $s \in \{2, 3, 5, 7, 9, 11, 13, 15\}$ on each subsample. Figure 8 reports end-to-end time (top row) and memory usage (bottom row) as the ratio grows. On every graph SPIN*'s curves are ordered cleanly by ratio at every s in both metrics: denser subsamples take longer and use more memory, with no inversions, indicating that the static-CPI pipeline scales smoothly with input size from sparse collaboration graphs to dense social graphs. The densest graph, `com-orkut`, also exposes the natural ceiling of vertex-centric $(1, s)$-nucleus decomposition at high s: the 100% curve cuts off near s=5 because the encoded clique count exceeds what a 30-minute budget can peel.

Exp-7: hierarchy emission against the level-DSU baseline. BuildHier (Section 7) consumes the preserved CPI once peeling finishes; the only available $(1, s)$ baseline that produces the same hierarchy is CND's general (r, s) level-DSU hierarchy specialised to r=1, which carries one parallel disjoint-set forest per core level (denoted K in the $O(K \cdot \Sigma)$ bound below) during the peel. We compare four algorithm variants on four representative graphs (Figure 9), sweeping $s \in \{3, 5, 7, 9, 11, 13, 15\}$: SPIN* (no hierarchy), SPIN*+BuildHier (our $O(\Sigma \log \Sigma)$ post-peel emit), CND (no hierarchy), and CND+**Hier** (level-DSU hierarchy during peel). All four solve the same problem; the comparison isolates the cost of hierarchy emission and of the underlying peel strategy. Three observations follow: (i) BuildHier adds at most 1.5× memory on top of SPIN*'s peel state (blue and green curves nearly overlap on every panel), while CND+Hier uses up to 43× more memory than SPIN*+BuildHier on matched cells (gmean 13× over 24 cells; orange far above green); (ii) even *without* emitting a hierarchy, CND (red) already uses more memory than SPIN*+BuildHier (green), and on `wiki-Talk` at $s \geq 9$ both CND variants exceed the 503 GB budget and never start, while ours finishes the entire sweep; (iii) the hierarchy emit is also cheaper at the larger-s cells where the witness mass dominates: BuildHier is up to 35× faster than CND+Hier's union work (gmean 7.7× over the 24 cells); on the smallest-s cells where the hierarchy emit is already sub-second the two are comparable, consistent with the $O(\Sigma \log \Sigma)$ vs. $O(K \cdot \Sigma)$ bounds.

Limitations. The static-CPI invariant is specific to r=1. For general (r, s) with $r \geq 2$, edge or higher-order peel events can destroy the cliques that a path stores, and the present pipeline does not subsume the mutable-CPI machinery of [4]. Memory is the binding resource on the densest graphs: Σ grows superpolynomially with s on dense inputs, so on a fixed-memory machine the largest

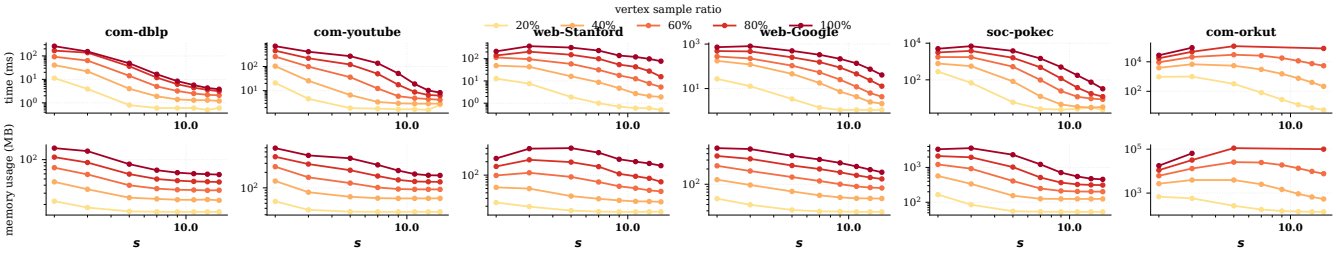

Figure 8: Vertex-induced scalability: time (top) and memory usage (bottom).

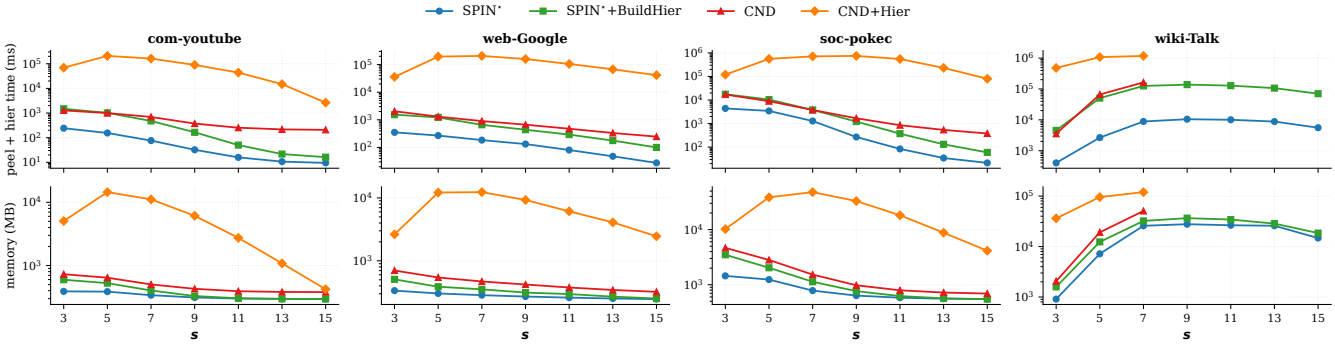

Figure 9: Hierarchy bench on four graphs: wall time (top) and peak memory (bottom).

reachable s is determined by the available RAM rather than by an algorithmic limit. Support counters use $\binom{p}{\eta}$ which exceeds 2^{63} at moderate p; the implementation guards the threshold pass with 128-bit accumulators when the count overflows 64 bits, but applications that need exact support magnitudes (rather than ranks) must use the same guard. Finally, the experiments here run single-machine; distributed-memory deployments and dynamic-graph updates are not evaluated.

Reproducibility. All experiments are scripted: build commands, dataset URLs, and bench drivers are version-controlled alongside the source. Each binary run is wrapped with `/usr/bin/time -v` so memory usage, page faults, and exit status are captured in every CSV row.

Takeaway. SPIN* is faster and lighter than CND on the 307 matched cells, the construction phase emerges as the new dominant cost once peeling is counter-only, ParaBuild addresses that shift, and SPIN confirms the direct fixed-point view's role as semantic reference rather than practical solver.

9 Case Studies

The experiments quantify cost; the case studies establish that the $r = 1$ specialisation is worth optimising in the first place. On the ground-truth tasks reported here, $(1, s)$ matches the quality of higher-r nuclei on the evaluated metrics at much lower cost, a claim scoped to the datasets and metrics below.

Setup. For cross-r comparisons we use the standard (r, s)-nucleus parameterisation [40]; the paper's algorithmic results apply only to $r = 1$, while the $r = 2$ and $r = 3$ scores serve as baselines from the same family. Because edge- and triangle-level scores need to be compared with vertex rankings, we project them by assigning each vertex the maximum score of an incident r-clique, then rank descending with ties broken uniformly.

Table 4: Active set on `com-dblp` as s grows.

s	nonzero $\lvert V\rvert$	% of n	max core	top-100 min	top-1k min
2	317,080	100.00	1.13×10^{2}	1.13×10^{2}	3.10×10^{1}
3	269,181	84.89	6.33×10^{3}	6.33×10^{3}	4.65×10^{2}
4	194,957	61.49	2.34×10^{5}	2.34×10^{5}	4.50×10^{3}
5	127,805	40.31	6.44×10^{6}	6.44×10^{6}	3.15×10^{4}
8	36,412	11.48	3.86×10^{10}	3.86×10^{10}	2.63×10^{6}
10	19,354	6.10	5.97×10^{12}	5.97×10^{12}	2.02×10^{7}
15	6,767	2.13	2.74×10^{17}	2.74×10^{17}	2.65×10^{8}
20	3,396	1.07	1.69×10^{21}	1.69×10^{21}	1.41×10^{8}
25	2,069	0.65	2.18×10^{24}	2.18×10^{24}	2.63×10^{6}
30	1,358	0.43	7.61×10^{26}	7.61×10^{26}	4.65×10^{2}

The ground-truth datasets are `com-dblp` and `com-youtube` from SNAP [49]. `com-dblp` has 317,080 vertices, 1,049,866 edges, and 5,000 venue-based ground-truth communities. `com-youtube` has 1,134,890 vertices, 2,987,624 edges, and 5,000 ground-truth communities.

We use three ranking metrics. Precision@K is the fraction of top-K vertices that belong to at least one ground-truth community. rec50 is the number of communities with at least 50% of their vertices inside the top-K. Triangle density is the number of triangles per vertex in the induced top-K subgraph.

Case study 1: s is a granularity knob. The first question is what s changes when $r = 1$. On `com-dblp`, increasing s sharply shrinks the active set while preserving the very top of the ranking. Table 4 reports the active-set size and top-prefix thresholds.

The active set shrinks from all vertices at $s = 2$ to 1,358 vertices at $s = 30$. This is a 234× reduction. At the same time, the maximum

Table 5: Ground-truth quality on com-dblp.

Method	top-K	Precision	rec50	tri/v
k-core	10,000	0.504	94	112.3
$(1, 3)$-core	10,000	0.523	114	113.6
$(1, 5)$-core	10,000	0.523	114	113.5
$(1, 10)$-core	10,000	0.523	114	113.4
$(1, 15)$-core	6,767	**0.548**	65	**154**
$(1, 30)$-core	1,358	0.507	4	**523**

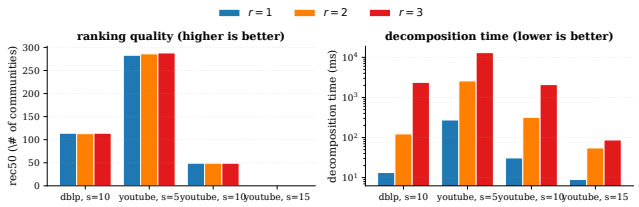

Figure 10: Cross-r comparison on com-dblp (s=10) and com-youtube ($s \in \{5, 10, 15\}$). Left: ranking quality (rec50, higher is better) is nearly identical across r. Right: decomposition time (log scale, lower is better) is 1–3 orders of magnitude smaller at r=1.

core value spans many orders of magnitude because high-s supports are binomial counts. The practical interpretation is simple. s controls how aggressively the ranking filters the graph.

Case study 2: $(1, s)$ improves the k-core ranking. On com-dblp, the first non-trivial clique support already improves the ground-truth ranking over k-core. Table 5 reports top-10,000 quality.

$(1, 3)$-core raises precision from 0.504 to 0.523 and rec50 from 94 to 114. For $s \in \{3, 5, 10\}$, the top-10,000 quality is identical in this table. For larger s, the active set drops below 10,000, so the comparison changes from ranking quality at fixed K to density of the surviving subgraph. $(1, 30)$-core reaches 523 triangles per vertex on 1,358 vertices.

Case study 3: higher r does not buy quality on the studied graphs. Edge- and triangle-level nuclei sit between $(1, s)$ and explicit clique enumeration on the cost axis; whether they buy quality at that cost is an open question on the same datasets. Fixing s and comparing $r \in \{1, 2, 3\}$ across com-dblp (s=10) and com-youtube ($s \in \{5, 10, 15\}$), Figure 10 shows the answer at a glance: within each cell the three quality bars are nearly equal heights (left panel), while the time bars span 1–3 orders of magnitude (right panel, log scale). On com-dblp at s=10, $(1, 10)$ and $(3, 10)$ agree on rec50 (114 vs. 114) while r=1 is 178× faster (13 ms vs. 2380 ms); on com-youtube the same pattern repeats for every s, with r=1 within 5 rec50 points of r=3 (at s=5: 283 vs. 288; at s=10 and s=15: identical) and between 9.8× and 68× faster than r=3 (and 6.2–10.3× faster than r=2). r=1 therefore dominates $r \in \{2, 3\}$ on the quality–runtime tradeoff for vertex ranking on these graphs; this does not prove higher r is never useful, only that increasing r here mostly adds cost.[2]

Case study 4: s distils a neighbourhood into near-clique modules. The previous studies use global rankings. The ego-network visualisation tests whether the same s knob also controls local interpretability. Figure 11 draws the two-hop ego-network of a com-dblp anchor on a shared spring layout under four successive

[2]At com-youtube s=15 the active set has 95 vertices, below the smallest SNAP community size, so rec50 is zero for every method.

Table 6: Nucleus hierarchy summary.

	$s = 3$			$s = 5$			$s = 10$			$s = 15$		
graph	br.	int.	d	br.	int.	d	br.	int.	d	br.	int.	d
com-dblp	10,900	426	4	6,050	278	4	751	20	4	197	8	4
com-youtube	13,507	353	4	714	4	3	12	1	2	1	0	1
web-Stanford	3,455	181	5	1,093	20	3	228	7	3	90	2	2

$(1, s)$-nuclei (s = 3, 5, 7, 10). A vertex of the ego is considered alive at s iff it participates in at least one s-clique entirely within the ego subgraph, so every alive vertex at s sits in a real K_s of the local $(1, s)$-nucleus. The four largest maximal cliques fully contained in the ego ($K_{31}, K_{25}, K_{19}, K_{18}$) are highlighted in colour with callouts.

These four cliques are alive at every s in $\{3, 5, 7, 10\}$ (their sizes are all at least 10), so they form a backbone that persists across all four panels. What changes from panel to panel is the peripheral alive set: vertices alive only because of smaller cliques peel off as s grows. Alive vertex count shrinks monotonically (1,356 → 721 → 370 → 165 out of 1,452): from "most of the neighbourhood is in some triangle" at s=3, to "only the four backbone cliques and their immediate clique-shared periphery survive" at s=10. The ego thus distils from a coauthor neighbourhood into a few large, tightly-coordinated coauthor communities — the same cliques are visible throughout, but the larger s strips away the loose periphery and isolates them.

Case study 5: hierarchy profiles differ by domain. BuildHier turns the core values into a $(1, s)$-clique-connected nucleus hierarchy (Definition 2.3). This makes it possible to compare how quickly the nucleus hierarchy collapses across domains. Table 6 reports the three hierarchy statistics used here. Figure 12 visualizes the resulting join trees side by side.

com-youtube collapses quickly: from 13,507 branches at s = 3 to a single branch at s = 15. com-dblp keeps the largest branch counts and a constant maximum depth across s, reflecting strong nested clique structure. web-Stanford retains nontrivial branches at every clique size, including s = 15. Figure 12 makes the contrast visual at s = 5 by plotting branch persistence ($k_{\text{birth}} - k_{\text{death}}$) against the rank of each branch when sorted by persistence (log–log). com-dblp's curve is a long flat tail: thousands of branches retain persistence above 10^2, the signature of a deep nested hierarchy. com-youtube's curve drops sharply after the first few ranks and plateaus near persistence 1, showing that only a handful of branches carry real depth. web-Stanford lies between the two. The hierarchy is therefore not only an output format. It gives a domain-level fingerprint of clique-witness nesting.

Takeaway. On the DBLP and YouTube ground-truth tasks, r = 1 matches higher-r quality metrics at much lower cost; the hierarchy and ego-network studies show the same decomposition also yields interpretable local structure and a per-domain fingerprint. The evidence is scoped to these datasets and metrics rather than a universal claim, but it is enough to justify optimising the exact r = 1 case.

10 Related Work

This paper uses prior ideas on core peeling, compact clique representations, local fixed-point characterisations, parallel construction, and hierarchy output; the new ingredient is the static-CPI invariant for r = 1.

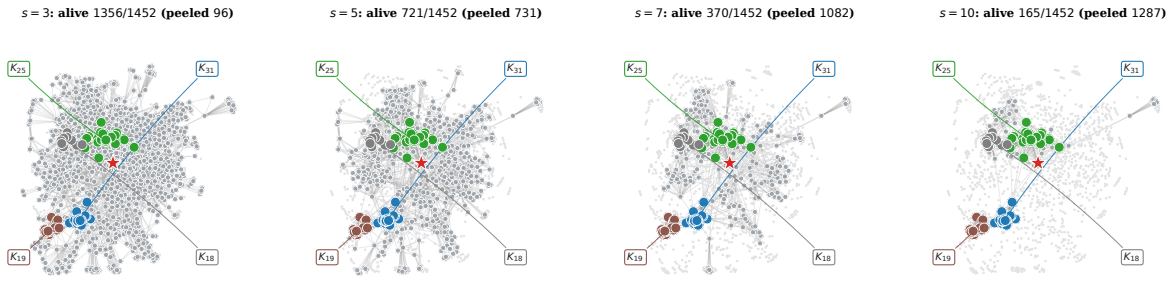

Figure 11: Same ego-network under four locally-restricted $(1, s)$-nuclei ($s = 3, 5, 7, 10$) on a shared spring layout. The four largest maximal cliques fully contained in the ego ($K_{31}, K_{25}, K_{19}, K_{18}$) form a backbone visible in every panel. As s grows the surrounding alive periphery shrinks from 1356 → 721 → 370 → 165 out of 1452, while the backbone cliques themselves remain intact — the local $(1, s)$-nucleus distils a loose coauthor neighbourhood into a few tight coauthor communities.

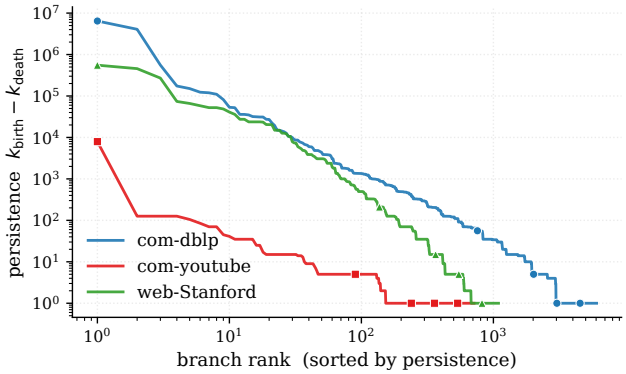

Figure 12: Branch persistence vs. rank at s=5 (log–log): long tail = deep hierarchy.

Enumeration-based core and nucleus decomposition. The k-core was introduced by Seidman [42] and is supported by a long line of efficient algorithms [6, 7, 9, 26, 36, 37, 46–48, 50]. Higher-order extensions strengthen the edge witness: k-truss [1, 11, 13, 23] uses triangle support, and *nucleus decomposition* [40, 41] generalises the witness to arbitrary s-cliques by peeling r-cliques according to their support in s-cliques. Hypergraph generalisations adopt the same peeling paradigm under different cohesion definitions [5, 8, 27, 30]. We specialise to the vertex-centric $r = 1$ case and exploit a state invariant that does not hold for general r.

Compact clique representations. Bron–Kerbosch search [10], Tomita pivoting [45], and degeneracy ordering [16] are standard tools for clique enumeration; Pivoter's succinct clique tree compactly encodes clique counts [24], and the Clique Path Index extends this line to support general nucleus decomposition with mutating hold/pivot updates during peeling [4]. The static-CPI counter form of this paper shows that for vertex peeling these path edits never need to be materialised.

Local and parallel peeling. h-index characterisations of core values [22] were studied for local core computation [32, 39] and extended to distributed and parallel settings [36, 50]. SPIN adapts that fixed-point view to the implicit s-clique witnesses encoded by the CPI, and SPIN* is its peel-order optimisation. The difference from prior local h-index work is not the operator but the representation: the witness multiset is held in static CPI paths and updated by binomial support-loss formulas.

Parallel construction and clique counting. Parallel clique search typically distributes the recursive enumeration work; ParaBuild follows the same seed-level decomposition. Because the downstream peel never mutates path sets, construction partitions into thread-independent subproblems whose results combine by a deterministic merge — an engineering consequence of the $r = 1$ state model rather than a new clique-enumeration paradigm.

Hierarchical cohesive subgraphs. Nucleus decompositions naturally form hierarchies across core levels [38, 40, 41, 43, 44], and elder-rule join-tree extraction is standard in computational topology [15]. Prior nucleus pipelines extract such hierarchies by sorting the decomposition output or revisiting clique witnesses; BuildHier instead reuses the preserved static CPI, treating each leaf as a hyperedge born at its minimum core value, so the hierarchy is a byproduct of preserving the CPI through peeling.

Applications of k-core and nucleus decomposition. Core and nucleus decompositions are workhorses for community detection [14, 18, 19, 21], densest-subgraph discovery [20], influential-vertex and role analysis [12, 25, 31], network visualization [51], and downstream uses in social analysis [3, 17], biological networks [29, 34], complex-network analysis [28], anomaly and security applications [2, 35], and recommendation [33]. This paper inherits the same downstream uses; the contribution is making the vertex-centric $(1, s)$ output cheaper to obtain at higher s.

Empirical graph mining methodology. The evaluation follows common practice in cohesive-subgraph experiments: SNAP community ground truth, large public graphs, runtime and memory comparisons, and dense synthetic stress tests [4, 41, 49].

11 Conclusion

For $(1, s)$-nucleus decomposition, vertex peeling never deletes edges, so the path structure of the CPI stays valid throughout the peel and exact decomposition reduces to counter maintenance over a static symbolic index. SPIN states this fixed-point, SPIN* realises it in deletion order, ParaBuild parallelises the construction that is now the bottleneck, and BuildHier recovers the nucleus hierarchy from

the static index. The matched configurations agree with the mutable baseline while reducing runtime and memory on the evaluated benchmarks, and the case studies show that vertex-centric $(1, s)$ matches higher-r quality metrics on community tasks at a fraction of the cost; the static-CPI invariant is specific to r=1, and extending it to restricted $r \geq 2$ settings, to streaming and dynamic graphs, and to GPU/distributed execution are the natural next steps.

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